Advanced Mathematical Deconstruction: ANOVA vs. MANOVA

Statistical Analysis Breakdown

1 Data Setup and Centroids

Let sample size $n_1 = n_2 = 3$, total $N = 6$, tracking $p = 2$ dependent variables over $k = 2$ groups.

$$\mathbf{Y}_{\text{Control}} = \begin{bmatrix} 4 & 3 \\ 6 & 5 \\ 5 & 4 \end{bmatrix}, \quad \mathbf{Y}_{\text{Trained}} = \begin{bmatrix} 8 & 7 \\ 10 & 9 \\ 9 & 8 \end{bmatrix}$$

The group mean vectors ($\bar{\mathbf{y}}_i$) and grand mean vector ($\bar{\mathbf{y}}$) calculate as:

$$\bar{\mathbf{y}}_1 = \begin{bmatrix} 5 \\ 4 \end{bmatrix}, \quad \bar{\mathbf{y}}_2 = \begin{bmatrix} 9 \\ 8 \end{bmatrix}, \quad \bar{\mathbf{y}} = \begin{bmatrix} 7 \\ 6 \end{bmatrix}$$

2 The ANOVA Approach (Scalar Partitioning)

ANOVA treats each variable independently, computing Sum of Squares (SS):

$$SS_{\text{Model}(Y_1)} = \sum_{i=1}^{2} n_i(\bar{y}_{i1} - \bar{y}_1)^2 = 3(5-7)^2 + 3(9-7)^2 = 24$$

$$SS_{\text{Error}(Y_1)} = \sum_{j=1}^{3}(y_{1j1} - 5)^2 + \sum_{j=1}^{3}(y_{2j1} - 9)^2 = 2 + 2 = 4$$

This yields an F-statistic of:

$$F = \frac{MS_{\text{Model}}}{MS_{\text{Error}}} = \frac{24/1}{4/4} = 24$$

Identical calculations for Y_2 yield $SS_{\text{Model}(Y_2)} = 24$ and $SS_{\text{Error}(Y_2)} = 4$.

3 The MANOVA Approach (Matrix Partitioning)

MANOVA retains the structural covariance by packaging calculations into $p \times p$ Sum of Squares and Cross-Products (SSCP) matrices.

3.1 Hypothesis (Between-Groups) Matrix H

$$\mathbf{H} = \sum_{i=1}^{2} n_i(\bar{\mathbf{y}}_i - \bar{\mathbf{y}})(\bar{\mathbf{y}}_i - \bar{\mathbf{y}})^T$$

$$\mathbf{H} = 3\begin{bmatrix} -2 \\ -2 \end{bmatrix}\begin{bmatrix} -2 & -2 \end{bmatrix} + 3\begin{bmatrix} 2 \\ 2 \end{bmatrix}\begin{bmatrix} 2 & 2 \end{bmatrix} = \begin{bmatrix} 24 & 24 \\ 24 & 24 \end{bmatrix}$$

3.2 Error (Within-Groups) Matrix E

Evaluating individual variation against group centroids:

$$\mathbf{E} = \sum_{i=1}^{2} \sum_{j=1}^{n_i} (\mathbf{y}_{ij} - \bar{\mathbf{y}}_i)(\mathbf{y}_{ij} - \bar{\mathbf{y}}_i)^T = \begin{bmatrix} 4 & 4 \\ 4 & 4 \end{bmatrix}$$

The total structural variation $\mathbf{T}$ preserves the fundamental matrix equality:

$$\mathbf{T} = \mathbf{H} + \mathbf{E} = \begin{bmatrix} 28 & 28 \\ 28 & 28 \end{bmatrix}$$

4 Multivariate Test Criteria Formulas

Significance is extracted using values determined from the matrix product $\mathbf{E}^{-1}\mathbf{H}$:

- **Wilks' Lambda:** $\Lambda = \frac{\det(\mathbf{E})}{\det(\mathbf{H}+\mathbf{E})}$
- **Pillai's Trace:** $V = \text{tr}(\mathbf{H}(\mathbf{H}+\mathbf{E})^{-1})$
- **Hotelling-Lawley Trace:** $\text{Trace} = \text{tr}(\mathbf{E}^{-1}\mathbf{H})$